MIL-R-11/3D
12 February 1968
SUPERSEDING
MIL-R-11/3C
17 April 1967

MILITARY SPECIFICATION

RESISTOR, FIXED, COMPOSITION (INSULATED),

STYLE RC20

* INACTIVE FOR DESIGN AFTER JANUARY 1968 (SEE 6.2) USE MIL-R-39008/2.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for style RC20 resistor.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-11 - Resistors, Fixed, Composition (Insulated), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

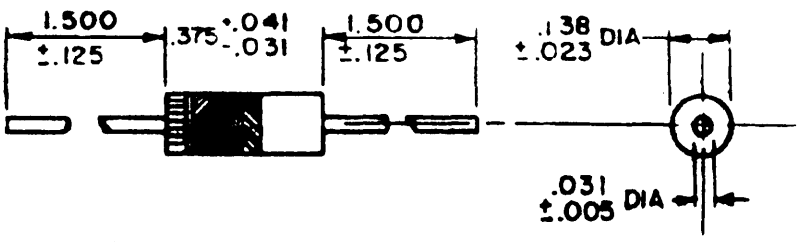
3.1 Requirements. Requirements shall be in accordance with MIL-R-11, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be 1/2 watt, based on full load operation at an ambient temperature of 70°C. Resistors operated at ambient temperatures in excess of 70°C shall be derated in accordance with figure 2. The rated direct current (dc) or root mean square (rms) continuous working voltage shall not exceed 350 volts.

3.4 Resistance. Minimum resistance value shall be 2.7 ohms; maximum resistance value shall be 22 megohms.

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INCHES	MM
.005	.13
.023	.58
.031	.79
.041	1.04
.125	3.18
.138	3.51
.375	9.53
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style RC20 resistor.

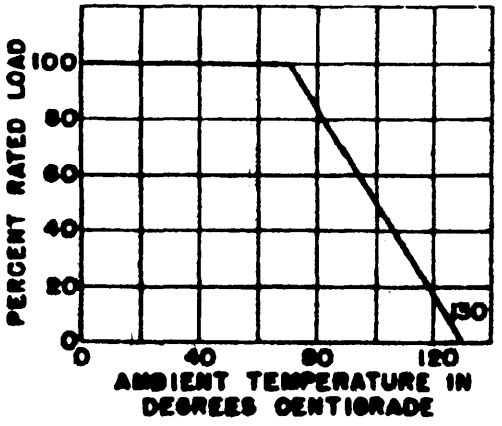


FIGURE 2. Derating curve for high ambient temperatures.

3.5 Moisture resistance. The change in resistance shall not exceed an average of 10 percent for each group of 10 resistors tested nor a maximum of 15 percent for any individual resistor.

3.6 Life. The change in resistance between the initial measurement and each of the succeeding measurements shall not exceed an average of 6 percent for each group of 10 resistors tested, nor a maximum of 10 percent for any individual resistor.

3.7 Resistance to soldering heat. The change in resistance shall not exceed ±(3 percent + 0.05 ohm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-11, and as specified herein.

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4.2 Terminal strength.

4.2.1 Direct load. Direct load shall be applied gradually until the load reaches 5 pounds.

4.2.2 Twist. The twist test is applicable to this specification.

4.3 Resistance to soldering heat. Temperature of solder in which resistor terminals are immersed, shall be 350° ±10°C.

4.4 Seal. The seal test is not applicable to this specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-11.

6. NOTES

6.1 The notes specified in MIL-R-11 are applicable to this specification.

* 6.2 Substitution data. The resistors specified herein are not for use in design after 12 February 1968. They are suitable for use in design contracts current to that date and to support existing Military equipment. Resistors specified in MIL-R-39008/2 are preferred for design and regardless of the failure rate designation can be used as substitutes for the inactivated resistors of the same resistance value and tolerance, of this specification.

* 6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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5. PROBLEM AREAS

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